⚙️ THE QUESTION THIS ANSWERS

Manager-Driven Turnover

Do people leave managers — not companies?

Tests the Gallup claim against a controlled synthetic dataset, using Cox survival analysis with a permutation test for manager clustering.

🔍 THE QUESTION THIS ANSWERS

Does turnover cluster by manager — and does that clustering survive controlling for grade and team size?

📐 THE APPROACH

Cox proportional hazards model on tenure, with grade and team size as covariates. Manager clustering tested via a residual-based permutation test over 500 shuffles — not directly encoded to avoid overfitting.

2,500	180	5yr	43%
employees	managers	observation window	observed exits

⚙️ WHAT I BUILT

Cox model, plus a permutation test.

It models how long someone stays before leaving — and properly accounts for the 57% who hadn't left yet, rather than dropping them.

```
survival        tenure_days
event           event_observed
covars          grade, team_size
manager         tested separately
```

🔴 TESTING THE MANAGER EFFECT

Shuffle who reported to whom 500 times, and check whether real teams differ more than random ones do.

01 Fit a baseline Cox model on grade and team size

02 Measure how far each person's exit defied the model (residuals)

03 Group by manager: how differently do their teams behave?

04 Shuffle manager labels 500× — what does chance alone look like?

05 Compare the real spread against those 500 random ones

⚠️ WHY THIS MATTERS

Grade and team size show no effect.

Neither covariate predicts who exits or when.

p = 0.57

Grade (1–5)

Seniority tells you nothing about who leaves

p = 0.72

Team size

Big teams and small teams lose people alike

0.51

Concordance (chance = 0.50)

No better than a coin flip at guessing who'll leave

The policy levers HR usually has on hand — grade bands, span of control — carry no signal in this dataset.

🔴 THE COST OF NOT KNOWING

Turnover clusters sharply by manager.

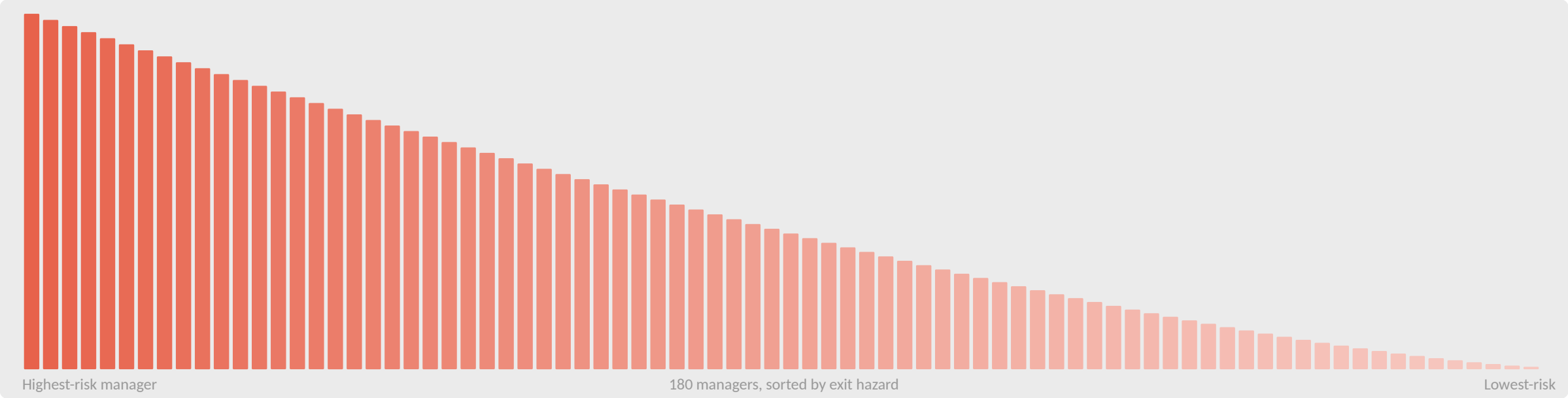
p < 0.002

Not one of 500 random shuffles came close

3.1×

More likely to lose people than the safest manager

Who you report to matters more to whether you stay than your job grade or the size of your team.



🔒 WHAT I BUILT

Sealed parameters caught a real bug.

The generator's true effect sizes were randomized, sealed, and not opened until the analysis was complete and committed to git.

STEP 01

p = 0.84

The first run said managers made no difference at all.

Permutation test on the baseline model's residuals.

STEP 02

Sealed file opened

The real answer, written down in advance, said managers mattered enormously.

True generating parameters, sealed until after the commit.

STEP 03

Bug found

A coding mistake matched every employee's result to the wrong manager.

Values reattached by row position instead of by employee ID.

STEP 04

p < 0.002

Matched correctly and rerun, the manager effect was unmistakable.

Both runs remain in the git history.

📊 WHAT I BUILT

Manager exit rates, filterable by grade.

The effect holds within individual grade bands, not just in aggregate.

TOOLS

PostgreSQL	DBeaver	Python · pandas
lifelines	Power BI	git

TAKEAWAY

If the goal is reducing turnover, manager quality is where to look — not grade or team-size policy levers, which showed no effect here.

Exit rate by manager

